

Project Presentation

Traffic law violation detection using AI techniques

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INTRODUCTION

- ▶ Problems of traffic law violations.
- ▶ Disadvantages of traditional enforcement methods.
- ▶ Use of automated violation detection.
- ▶ Improving road safety and reducing the number of accidents caused by negligent driving.

MOTIVATION

- ▶ High Rate of Traffic Violation.
- ▶ Manual Monitoring Limitations.
- ▶ Need for Automation in Law Enforcement.
- ▶ Scalability for Future Implementation.

OBJECTIVE

- ▶ Develop an Automated Violation Detection System.
- ▶ Leverage AI and Computer Vision Technologies.
- ▶ Capture and Store Violation Evidence.

METHODOLOGY

- ▶ **Dataset Collection:** Gathering videos of vehicles from roads.
- ▶ **Deep Learning Models:** Training models to detect vehicles turning without indicators.
- ▶ **Feature Extraction:** Using image processing techniques to identify relevant features for classification.
- ▶ **Model Training:** Utilizing neural networks to detect and classify violations.
- ▶ **Data Logging:** Storing detected violations for future analysis and reporting in folder.
- ▶ **Real-time Monitoring:** Implementing the process on live traffic footage.

TECHNOLOGIES USED

- ▶ **OpenCV:** For initial video processing
- ▶ **Ultralytics YOLO:** For detection of vehicles.
- ▶ **Kalman Filter:** Predicts and tracks vehicles
- ▶ **Deep Sort:** Keep track of same car
- ▶ **AKAZE:** Extract visual features, focused on turn signal area

TECHNOLOGIES USED

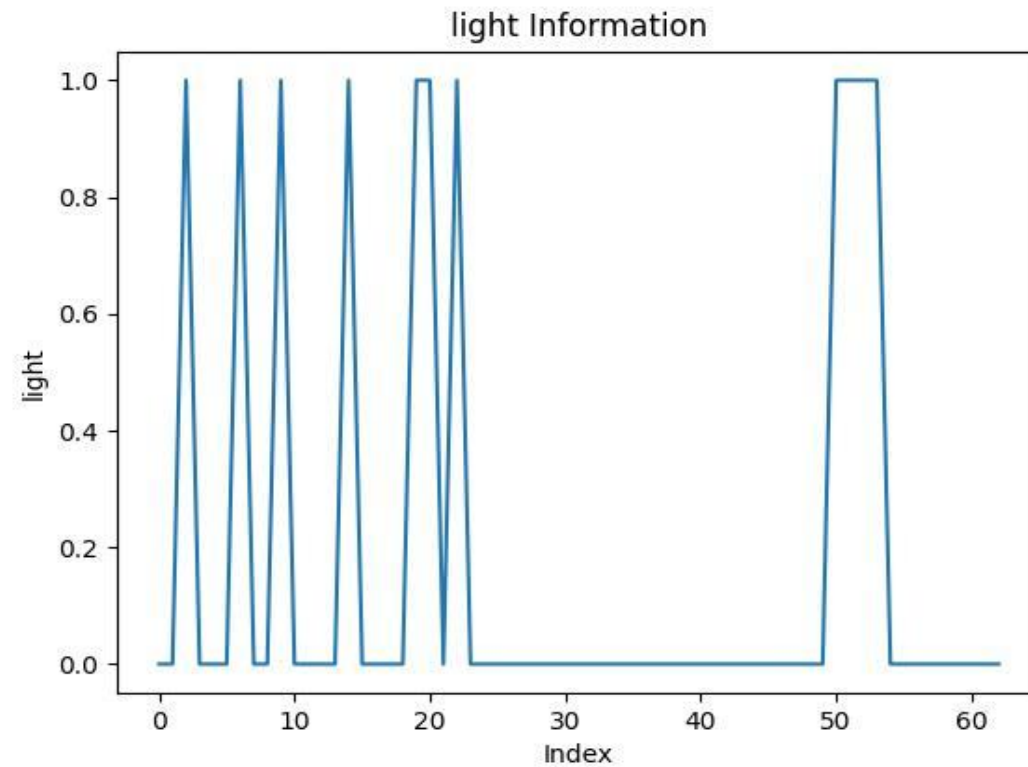
- ▶ **PyTorch:** Powers deep learning model, to identify vehicle are turning.
- ▶ **CSV:** Stores the processed data, include vehicles IDs and tracks
- ▶ **Matplotlib:** Generates visualization, such as graphs

VEHICLE ANALYSIS PROCEDURE

- ▶ **Assigns IDs to each vehicles:** YOLO assigns IDs to each vehicle.
- ▶ **Finding their trajectories:** Deep Sort & Kalman filter tracks vehicles.
- ▶ **Detecting turn vehicles:** Kalman filter based vehicle turn detection.
- ▶ **Detecting law violating turns:** Combining YOLO and AKAZE to identify vehicles and their turning maneuvers with analysis of light patterns.

Indicator Detection

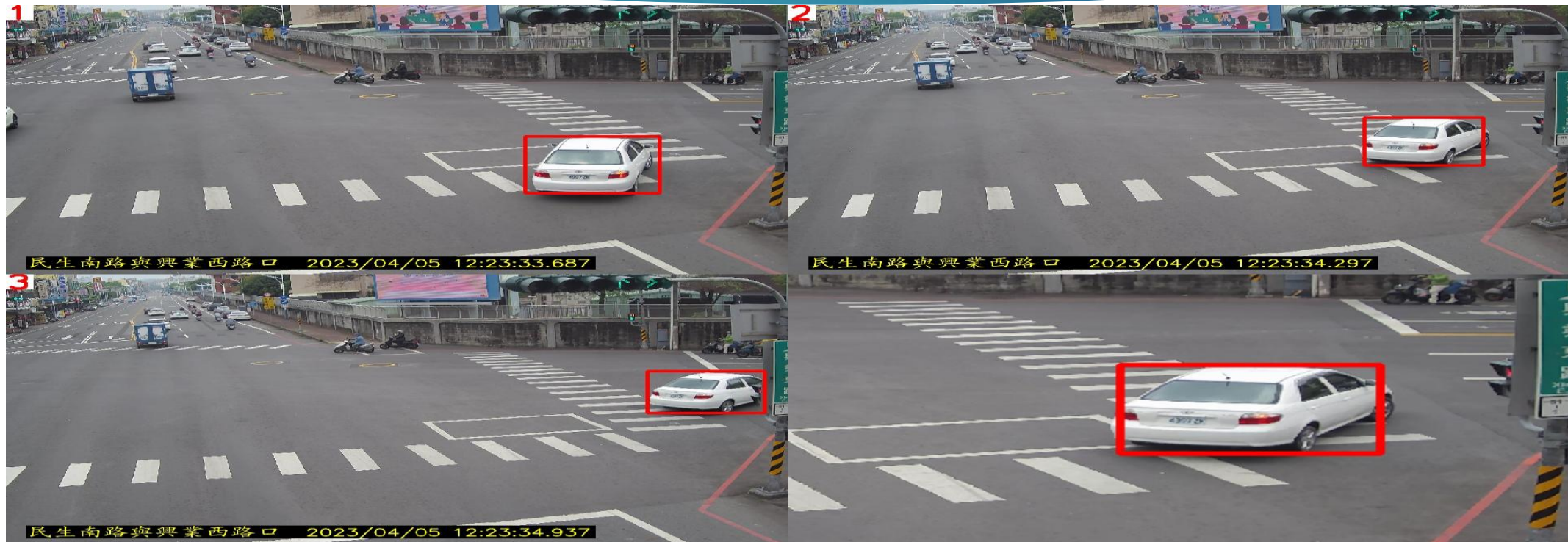
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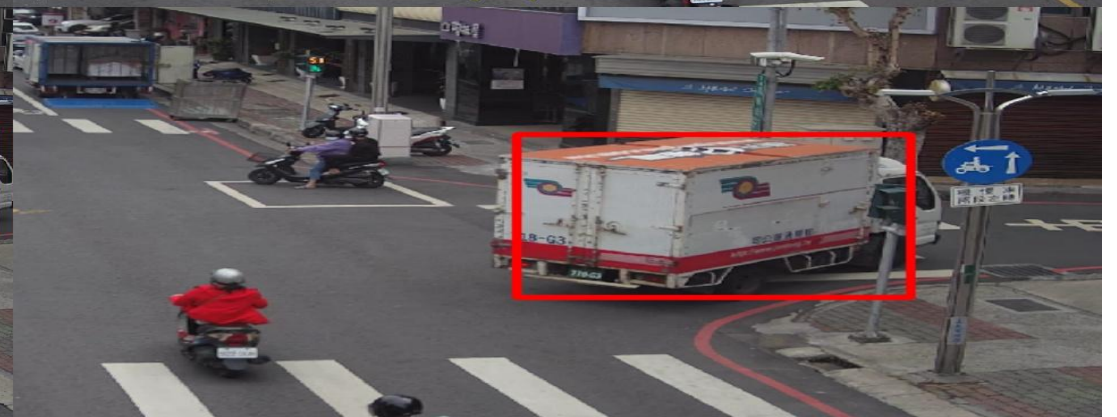
Turn Detection

	A	B	C
1	video_name	car_id	predict_turn
2	intersection_surveillance	7	straight
3	intersection_surveillance	47	right
4	intersection_surveillance	1	straight
5	intersection_surveillance	37	straight
6			

Vehicle way tracking (Car)



Vehicle way tracking (Truck)



CONCLUSION

- ▶ The system effectively integrates YOLO, Kalman filters, AKAZE and other AI techniques to automate vehicle tracking and turn detection
- ▶ The automated approach enhances traffic management.

FUTURE SCOPE

- ▶ Improve systems ability to handle large volumes of video data and real world challenges
- ▶ Further increase in stability and accuracy

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